Supplement to "Maximum Blackouts That Keep Every Lattice Rectangle Identifiable"

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This supplement collects the computed values, the evidence for each computer-assisted statement, and the details of the certificates and formal checks. Numbers of theorems, lemmas, and sections refer to the paper. All paths start from `kagey-problems/problem001` in the repository cited there.

S1 Computed values

grid	f	$a_h(m)$	by
2×2	4	1	HIB
2×3	4	12	HIB
2×4	5	32	HIB
2×5	6	80	IB
2×6	7	192	TI
2×7	8	448	TI
3×3	5	53	IB
3×4	6	224	IB
3×5	7	892	IB
3×6	8	3,420	IB
3×7	9	12,704	IB
3×8	10	45,864	IB
3×9	11	161,992	IB
3×10	12	562,912	ITD

grid	f	$a_h(m)$	by
4×4	7	316	IB
4×5	8	1,810	IB
4×6	10	26	IB
4×7	11	292	IB
4×8	12	2,118	IMD
5×5	10	8	IB
5×6	11	76	ISC
5×7	12	1,184	ISC
5×8	14	4	TSDC
5×9	15	44	TSC
5×10	16	350	TSC
5×11	17	3,004	TSC
5×12	18	24,706	TSCM
6×6	12	160	ISC

grid	f	$a_h(m)$	by
6×7	13	2,536	SC
6×8	15	8	SC
6×9	16	–	TS
6×10	17	–	TS
6×11	18	–	TS
6×12	20	–	TS
6×13	21	–	TS
6×14	22	–	TS
6×15	24	–	TS
7×7	14	7,300	SC
7×8	16	≥ 18	S
7×24	36	–	T
7×25	37	–	T
7×26	38	–	T
7×27	39	–	T

Table S1: Computed values for $h \leq m$. H: hand, T: theorem, I: ILP (value only on 3×10, 4×8, 5×6, 5×7, and 6×6), B: brute force, D: equality candidates, C: column search, S: two SAT encodings, with complete enumeration for exact counts, M: transfer counter, –: not established. On 7×8 only a lower bound for the count is given.

As a check on rectangle generation, the totals match A085582 for squares and A289832 for rectangular grids. The constructions in `jis/data/table_counts/sat_7x8_16.json` give $f(7,8) \geq 16$, completing $f(7,8) = 16$. The separate enumeration `jis/data/table_counts/sat2split_7x8_16.json` gives 18 valid constructions with 16 points. This proves only $a_7(8) \geq 18$.

S2 Evidence for the computer-assisted statements

statement	method; independent check	evidence
Cor. 32, Thm. 46(b)	candidates; brute force	`verify_narrow.py`
Thms. 25, 26, Rem. 30	SymPy; transfer matrix	`three_row_*.json`
Thm. 37, Cor. 38, Table 1	Prop. 36; candidates	`four_row_*.json`
Thms. 40(a), 45	encoding A, LRAT; encoding B	`unsat_certificates.json`, `small_cells_*`
Thms. 40(b), 43(c)	collar board; enumerators, Lean	`five_row_*family*`, `six_row_*w15*`
Thm. 40(c), Rem. 41	SAT, counter; candidate search, second counter	`five_row_*`
Thm. 43, upper bounds	structured, LRAT; second encoder, search	`unsat_certificates.json`, `six_row_sat_w*`, `six_row_excess8_*`
Thm. 43(a), lower bound	collar board; two enumerators	`six_row_collar_6x9_check.json`
Thm. 43(b), lower bound	three enumerators	`six_row_excess8_family.json`
Thm. 44	structured SAT, collar board; enumerators, brute force	`seven_row_sat_w*`, `seven_row_witness_brute.json`
Thm. 47	hand proof, Lean; full boards for $h = 5, \ldots, 9$ from the definition	`general_theorem_board_check.txt`, `GeneralTheorem.lean`
Table S1	SAT; column search	`table_counts/*`

Table S2: Computer-assisted statements, their method and check, and the evidence. Evidence files are in `jis/data`, and `verify_narrow.py` is in the problem folder.

S3 Certificates

The CNF files for the cell instances and the 70 structured instances with $m + 9$ points are written twice, by two independently written generators, and the clauses agree. We split 6×8 by its first column into 64 cases, and 7×7 and 7×8 into 128 each. The second generator checks that these splits exhaust the first-column choices. The 345 instances with $m + 8$ points on six rows, one for each labeled graph of Lemma 42 at widths 9, 10, and 11, are written by `jis/scripts/six_row_excess8_export_cnf.py`. Encoding B, run separately on the same 345 cases, also finds every one unsatisfiable. The hashes of the CNF files, the proofs, and the results of both proof checkers are in `jis/data/unsat_certificates.json` and `jis/data/six_row_excess8_certificates.json`. The proofs themselves are too large for the repository.

S4 Formal checks

Kernel-checked validity tests cover S_0, S_8, the construction in Theorem 45, the 6×15 construction in Theorem 43, and the 5×13 and 6×23 collar boards. The formal proofs also show that at most one rectangle can be hidden. The kernel also checks that $\{(1,2),(2,1),(3,2),(2,3)\}$ hides $D(1,1)$ on 4×3. Thus the restriction $m \geq 4$ in Theorem 33 is necessary.

The remaining unformalized parts are Lemma 13, Theorem 11, Lemma 42, $f(5,5) \leq 10$, Lemma 54, the seven-row constructions, and the other results in Sections 3 to 10, except Section 8, which is formalized in full (`GeneralLocal.lean` and `GeneralTheorem.lean`).